Tutorial-9: Black-Box & White-Box Testing

Module Name: Flipkart Web Application - Functional, UI, Integration & System Test Cases

Test Case ID st_09
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Test Case D: To validate Black-Box, White-Box, Integration and System Testing on Flipkart e-commerce web application (https://www.flipkart.com)
Prerequisite: 1. Stable internet connection 2. Modern browser (Chrome/Edge) with DevTools 3. Valid Flipkart URL 4. Test user credentials (optional)
Environmen: OS: Windows 11 / macOS | Browser: Google Chrome | Webpage URL: https://www.flipkart.com

Scenario ID	Scenario	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
BB-1a	Homepage loads successfully	1. Open https://www.flipkart.com 2. Observe page load	Valid URL	Homepage loads with header, search bar, categories and banners	Homepage loaded correctly with all major UI elements visible	Pass	Black-box functional test - positive flow
BB-1b	Search for a product (True Wireless earbuds)	1. Click on search bar 2. Type 'True Wireless' 3. Press Enter / click search	Search term: True Wireless	Search results page showing relevant products with filters	Results page displayed with 16,363 products and left-side filters	Pass	Black-box search functionality
BB-1c	Apply category filter - True Wireless	1. On results page go to left Filters 2. Navigate Categories → Audio & Video → Headset → Earphones → Wireless Earphones → True Wireless	Category navigation path	Only True Wireless products shown; breadcrumb updates	Category filter applied correctly; breadcrumb: Home > Audio & Video > ... > True Wireless	Pass	Black-box filter/category navigation
BB-1d	Sort products by Popularity	1. On results page click Sort By → Popularity	Sort option: Popularity	Products re-ordered by popularity	Products sorted by Popularity (default selected)	Pass	Black-box sorting
BB-1e	Open product detail page	1. Click on any product (e.g. TECHZONE Premium Bluetooth Neckband) 2. Observe product detail page	Click product card	Product detail page opens with images, price, offers, Add to Cart & Buy Now	Product page opened successfully showing price ₹242, offers and action buttons	Pass	Black-box product detail view
BB-1f	Price & discount display validation	1. On product page check original price, discounted price and % off	Product with discount	Correct original price, discounted price and percentage off shown	Original ₹999, Discounted ₹242 (76% off) displayed correctly	Pass	Black-box price calculation display
BB-1g	Add to Cart / Buy Now button presence	1. On product page locate Add to Cart and Buy Now buttons	None	Both buttons visible and clickable	Add to Cart (white) and Buy Now (yellow) buttons present	Pass	Black-box UI element check
BB-1h	Bank offers & delivery location section	1. On product page check Bank offers section 2. Check Delivery details section	None	Bank offers listed; delivery location selector present	Bank offers (Axis/SBI) and 'Location not set - Select delivery location' shown	Pass	Black-box offer & delivery UI
WB-2a	Console errors on Homepage load (White-Box)	1. Open Flipkart homepage 2. Open DevTools → Console tab 3. Observe errors	DevTools open	No critical CORS or resource load failures expected	Multiple CORS errors for static SVG assets (L1Nav/*.svg) and failed resource loads observed	Fail	White-box: CORS policy blocks x-user-agent header on preflight; net::ERR_FAILED for several SVGs
WB-2b	Console errors on Search/Listing page	1. Navigate to True Wireless listing 2. Open Console 3. Observe network & console errors	DevTools open on listing page	Clean console or only non-critical warnings	POST to sonic.fdp.api.flipkart.com returned 406 (Not Acceptable); another POST returned 500 Internal Server Error	Fail	White-box: API collector/business endpoint returning 406 & 500
WB-2c	Console errors on Product Detail page	1. Open any product detail page 2. Open Console 3. Note warnings & errors	DevTools open on PDP	Minimal warnings; no blocking errors	Deprecated apple-mobile-web-app-capable meta, missing RN components (Placeholder, RNShimmeringView etc.), CORS on codes.json, 406 on collector API	Fail	White-box: React Native Web component fallbacks + CORS + 406
WB-2d	Service Worker registration	1. On any page open Console 2. Check ServiceWorker messages	None	Service Worker registers successfully	ServiceWorker registration successful with scope https://www.flipkart.com/	Pass	White-box: Service Worker works correctly
IT-3a	Search → Filter → Product Detail integration	1. Search 'True Wireless' 2. Apply category filter 3. Click a product 4. Verify product matches filter	Search + filter path	Product opened belongs to True Wireless category	Product detail correctly linked from filtered listing	Pass	Integration of Search + Category Filter + PDP
IT-3b	Price display consistency across listing & PDP	1. Note price on listing card 2. Open product 3. Compare price on detail page	Same product	Price and discount % match between listing and detail	Listing showed ₹242 (75% off); PDP showed ₹242 (76% off) – minor % difference due to rounding	Pass	Integration of listing price service and PDP price service
IT-3c	Offers section integration with payment banks	1. On PDP expand Bank offers 2. Observe Axis / SBI offers	None	Offers from integrated bank partners displayed	Flipkart Axis and Flipkart SBI cashback offers shown with Apply buttons	Pass	Integration of Offer engine with Bank partners
ST-4a	Header & navigation consistency	1. Observe top header on homepage, listing and PDP	None	Flipkart logo, Search, Login, Cart consistent across pages	Header elements consistent on all tested pages	Pass	System-level UI consistency
ST-4b	Responsive layout of product cards	1. On listing page resize browser or use device toolbar	Different viewport widths	Product grid adapts; cards remain usable	Grid layout remains responsive	Pass	System UI responsiveness
ST-4c	Trending section & banners on homepage	1. Scroll homepage 2. Check Trending Gadgets & banners	None	Banners and trending products load without broken images	Healthcare, Infinix, Mivi banners and Trending Gadgets section loaded correctly	Pass	System content loading
ST-4d	Login button & Cart icon presence	1. On any page locate Login and Cart icons	None	Login dropdown and Cart icon always visible in header	Login and Cart present on homepage, listing and PDP	Pass	System navigation elements
ST-4e	Assured badge & rating display	1. On listing page check products with Assured badge and star ratings	None	Assured badge and rating stars shown for eligible products	Assured badge and ratings (e.g. 4.1★, 4.3★) displayed correctly	Pass	System trust signals

SUMMARY OF TEST RESULTS

Total Test C:	20
Passed	17
Failed	3
Pass %	85.00%

ISSUES FOUND (White-Box / Console)

Bug ID	Title	Severity	Description
BUG-WB-1	CORS errors on static SVG assets (L1Nav)	High	Multiple Access-Control-Allow-Headers violations for x-user-agent on static-assets-web.flixcart.com SVG icons
BUG-WB-2	API 406 Not Acceptable on collector/business endpoint	Medium	POST https://sonic.fdp.api.flipkart.com/4/data/collector/business returns 406
BUG-WB-3	Internal Server Error 500 on push/callback	Medium	POST https://1.rome.api.flipkart.com/api/1/connekt/push/callback returns 500
BUG-WB-4	Missing React Native Web components	Low	Placeholder, RNShimmeringView, GifView, LiveVideoView, HybridConditionTrackerView not available – fallback views used